Triad

Component Design Document

1 Description

Triad attitude guidance. Computes the reference attitude that aligns the body thrust direction with a requested inertial thrust direction while keeping the solar array drive axis orthogonal to the sun. Wraps the TriadAlgorithm C++ algorithm via its C shim.

2 Requirements

No requirements have been specified for this component.

3 Design

3.1 At a Glance

Below is a list of useful parameters and statistics that give a quick look into the makeup of the component.

- **Execution** - *passive*
- **Number of Connectors** - 5
- **Number of Invokee Connectors** - 3
- **Number of Invoker Connectors** - 2
- **Number of Generic Connectors** - *None*
- **Number of Generic Types** - *None*
- **Number of Unconstrained Arrayed Connectors** - *None*
- **Number of Commands** - *None*
- **Number of Parameters** - 3
- **Number of Events** - *None*
- **Number of Faults** - *None*
- **Number of Data Products** - 1
- **Number of Data Dependencies** - 2
- **Number of Packets** - *None*

3.2 Diagram

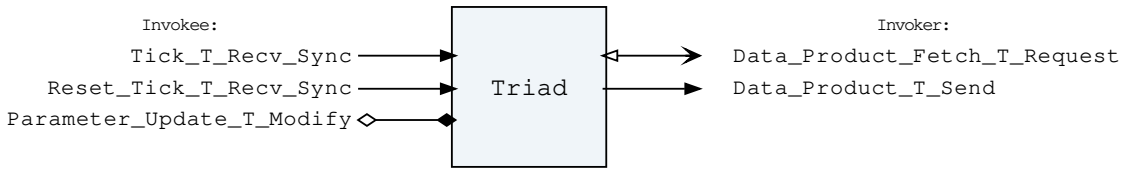


Figure 1: Triad component diagram.

3.3 Connectors

Below are tables listing the component's connectors.

3.3.1 Invokee Connectors

The following is a list of the component's *invokee* connectors:

Table 1: Triad Invokee Connectors

Name	Kind	Type	Return_Type	Count
Tick_T_Recv_Sync	recv_sync	Tick.T	-	1
Reset_Tick_T_ Recv_Sync	recv_sync	Tick.T	-	1
Parameter_ Update_T_Modify	modify	Parameter_ Update.T	-	1

Connector Descriptions:

- **Tick_T_Recv_Sync** - Run the algorithm up to the current time.
- **Reset_Tick_T_Recv_Sync** - Apply any pending parameter update. Called on GNC state change.
- **Parameter_Update_T_Modify** - The parameter update connector.

3.3.2 Invoker Connectors

The following is a list of the component's *invoker* connectors:

Table 2: Triad Invoker Connectors

Name	Kind	Type	Return_Type	Count
Data_Product_ Fetch_T_Request	request	Data_Product_ Fetch.T	Data_Product_ Return.T	1
Data_Product_T_ Send	send	Data_Product.T	-	1

Connector Descriptions:

- **Data_Product_Fetch_T_Request** - Fetch a data product item from the database.
- **Data_Product_T_Send** - The data product invoker connector

3.4 Interrupts

This component contains no interrupts.

3.5 Initialization

Below are details on how the component should be initialized in an assembly.

3.5.1 Component Instantiation

This component contains no instantiation parameters in its discriminant.

3.5.2 Component Base Initialization

This component contains no base class initialization, meaning there is no `init_Base` subprogram for this component.

3.5.3 Component Set ID Bases

This component contains commands, events, packets, faults, or data products that require a base identifier to be set at initialization. The `set_Id_Bases` procedure must be called with the following parameters:

Table 3: Triad Set Id Bases Parameters

Name	Type
`Data_Product_Id_Base`	`Data_Product_Types.Data_Product_Id_Base`
`Parameter_Id_Base`	`Parameter_Types.Parameter_Id_Base`

Parameter Descriptions:

- **`Data_Product_Id_Base`** - The value at which the component's data product identifiers begin.
- **`Parameter_Id_Base`** - The value at which the component's parameter identifiers begin.

3.5.4 Component Map Data Dependencies

This component contains data dependencies that need their IDs mapped to valid data product IDs at initialization. Each data dependency also needs a stale limit provided in microseconds. If the timestamp of the data dependency is older than the reference time (usually the current time) minus the stale limit then the data dependency is considered too old to safely use. The `map_Data_Dependencies` procedure must be called with the following parameters:

Table 4: Triad Map Data Dependencies Parameters

Name	Type
`Sun_Direction_Inertial_Id`	`Data_Product_Types.Data_Product_Id`
`Sun_Direction_Inertial_Stale_Limit`	`Ada.Real_Time.Time_Span`
`Thrust_Direction_Body_Id`	`Data_Product_Types.Data_Product_Id`
`Thrust_Direction_Body_Stale_Limit`	`Ada.Real_Time.Time_Span`

Parameter Descriptions:

- **`Sun_Direction_Inertial_Id`** - The external data product ID used to fetch the value for the 'Sun_Direction_Inertial' data dependency.
- **`Sun_Direction_Inertial_Stale_Limit`** - The stale limit used for the 'Sun_Direction_Inertial' data dependency. If the fetched data product is older than the given Time_Span then it is deemed stale and unfit for use by the component.
- **`Thrust_Direction_Body_Id`** - The external data product ID used to fetch the value for the 'Thrust_Direction_Body' data dependency.

- **`Thrust_Direction_Body_Stale_Limit`** - The stale limit used for the 'Thrust_Direction_Body' data dependency. If the fetched data product is older than the given Time_Span then it is deemed stale and unfit for use by the component.

3.5.5 Component Implementation Initialization

The calling of this implementation class initialization procedure is mandatory. Initializes the triad algorithm with the default parameter values. The `init` subprogram takes no parameters.

3.6 Commands

The Triad component has no commands.

3.7 Parameters

Parameters for the Triad component.

Table 5: Triad Parameters

Local ID	Parameter Name	Type	Default Value
0x0000 (0)	Sada_Hat_B	Packed_F32x3.T	[1.0,0.0,0.0]
0x0001 (1)	Thrust_Req_Hat_N	Packed_F32x3.T	[1.0,0.0,0.0]
0x0002 (2)	N3_Axis	Packed_N3_Axis.T	(Value=>Triad_Enums.N3_Axis.Plus_Z_Hat_N)

Parameter Descriptions:

- **`Sada_Hat_B`** - [-] Solar array drive axis in body frame components. Must be within 1e-3 of unit length.
- **`Thrust_Req_Hat_N`** - [-] Requested thrust direction in inertial frame components. Must be within 1e-3 of unit length.
- **`N3_Axis`** - [-] Inertial z axis direction used as the fallback constraint axis when the sun direction is nearly parallel to the requested thrust direction.

3.8 Events

The Triad component has no events.

3.9 Data Products

Data products for the Triad component.

Table 6: Triad Data Products

Local ID	Data Product Name	Type
0x0000 (0)	Attitude_Reference	Att_Ref.T

Data Product Descriptions:

- **`Attitude_Reference`** - Reference attitude that aligns the thrust direction with the requested inertial direction. The algorithm produces an attitude only, so the reference rate and acceleration are zero. The attitude is zero when the triad cannot be formed.

3.10 Data Dependencies

Data dependencies for the Triad component.

Table 7: Triad Data Dependencies

Data Dependency Name	Type
`Sun_Direction_Inertial`	`Packed_F32x3.T`
`Thrust_Direction_Body`	`Packed_F32x3.T`

Data Dependency Descriptions:

- **`Sun_Direction_Inertial`** - [-] Unit vector from the spacecraft to the sun in inertial frame components.
- **`Thrust_Direction_Body`** - [-] Unit thrust direction in body frame components.

3.11 Packets

The Triad component has no packets.

3.12 Faults

The Triad component has no faults.

4 Unit Tests

The following section describes the unit test suites written to test the component.

4.1 *Triad_ Tests* Test Suite

This is a unit test suite for the Triad component

Test Descriptions:

- **`Test`** - Run the algorithm on the input sets of the Python reference test to ensure the Ada to C to C++ integration is sound.
- **`Test_Degenerate_Geometry`** - Check the fallback constraint axis when the sun is parallel to the requested thrust direction, and the zero attitude when the triad cannot be formed.
- **`Test_Invalid_Parameter`** - Ensure a staged configuration the algorithm would reject is refused at validation.
- **`Test_Invalid_Data_Dependency`** - Ensure a data dependency with the wrong identifier is treated as a wiring defect and fails the tick's assertion.

5 Appendix

5.1 Preamble

This component contains no preamble code.

5.2 Packed Types

The following section outlines any complex data types used in the component in alphabetical order. This includes packed records and packed arrays that might be used as connector types, command arguments, event parameters, etc..

Att_Ref.T:

Table 8: Att_Ref Packed Record : 288 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Sigma_Rn	Packed_F32x3.T	-	96	0	95
Omega_Rn_N	Packed_F32x3.T	-	96	96	191
Domega_Rn_N	Packed_F32x3.T	-	96	192	287

Field Descriptions:

- **Sigma_Rn** - *No description provided.*
- **Omega_Rn_N** - *No description provided.*
- **Domega_Rn_N** - *No description provided.*

Data_Product.T:

Generic data product packet for holding arbitrary data types

Table 9: Data_Product Packed Record : 1096 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Data_Product_Header.T	-	72	0	71	-
Buffer	Data_Product_Types.Data_Product_Buffer_Type	-	1024	72	1095	Header.Buffer_Length

Field Descriptions:

- **Header** - The data product header
- **Buffer** - A buffer that contains the data product type

Data_Product_Fetch.T:

A packed record which holds information for a data product request.

Table 10: Data_Product_Fetch Packed Record : 16 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Data_Product_Types.Data_Product_Id	0 to 65535	16	0	15

Field Descriptions:

- **Id** - The data product identifier

Data_Product_Header.T:

Generic data_product packet for holding arbitrary data_product types

Table 11: Data_Product_Header Packed Record : 72 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time	Sys_Time.T	-	48	0	47
Id	Data_Product_Types.Data_Product_Id	0 to 65535	16	48	63
Buffer_Length	Data_Product_Types.Data_Product_Buffer_Length_Type	0 to 128	8	64	71

Field Descriptions:

- **Time** - The timestamp for the data product item.
- **Id** - The data product identifier
- **Buffer_Length** - The number of bytes used in the data product buffer

Data_Product_Return.T:

This record holds data returned from a data product fetch request.

Table 12: Data_Product_Return Packed Record : 1104 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
The_Status	Data_Product_Enums.Fetch_Status.E	0 => Success 1 => Not_Available 2 => Id_Out_Of_Range	8	0	7	-
The_Data_Product	Data_Product.T	-	1096	8	1103	-

Field Descriptions:

- **The_Status** - A status relating whether or not the data product fetch was successful or not.
- **The_Data_Product** - The data product item returned.

Packed_F32x3.T:

Packed 3 element vector of 32-bit floats.

Table 13: Packed_F32x3 Packed Array : 96 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Short_Float	-3.40282e+38 to 3.40282e+38	32	3	96

Packed_N3_Axis.T:

Triad fallback constraint axis selection, packed as an 8-bit enumeration. Values outside the enumeration fail type validation during parameter staging.

Table 14: Packed_N3_Axis Packed Record : 8 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Value	Triad_Enums.N3_Axis.E	0 => Plus_Z_Hat_N 1 => Minus_Z_Hat_N	8	0	7

Field Descriptions:

- **Value** - The fallback constraint axis.

Parameter.T:

Generic parameter packet for holding a generic parameter

Table 15: Parameter Packed Record : 792 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Parameter_Header.T	-	24	0	23	-
Buffer	Parameter_Types.Parameter_Buffer_Type	-	768	24	791	Header.Buffer_Length

Field Descriptions:

- **Header** - The parameter header
- **Buffer** - A buffer that contains the parameter type

Parameter_Header.T:

Generic parameter header for holding arbitrary parameters

Table 16: Parameter_Header Packed Record : 24 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Parameter_Types.Parameter_Id	0 to 65535	16	0	15
Buffer_Length	Parameter_Types.Parameter_Buffer_Length_Type	0 to 96	8	16	23

Field Descriptions:

- **Id** - The parameter identifier
- **Buffer_Length** - The number of bytes used in the parameter type buffer

Parameter_Update.T:

A record intended to be used as a provide/modify connector type for updating/fetching parameters.

Table 17: Parameter_Update Packed Record : 824 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Table_Id	Parameter_Types.Parameter_Table_Id	0 to 65535	16	0	15	-
Operation	Parameter_Enums.Parameter_Operation_Type.E	0 => Stage 1 => Update 2 => Fetch 3 => Validate 4 => Discard	8	16	23	-
Status	Parameter_Enums.Parameter_Update_Status.E	0 => Success 1 => Id_Error 2 => Validation_Error 3 => Length_Error	8	24	31	-
Param	Parameter.T	-	792	32	823	-

Field Descriptions:

- **Table_Id** - The ID for the table that contains this parameter
- **Operation** - The parameter operation to perform.
- **Status** - The parameter return status.
- **Param** - The parameter that has been updated or fetched.

Sys_Time.T:

A record which holds a time stamp using EMA VTC format including seconds and subseconds since epoch (1-5-1980 to 1-6-1980 midnight).

Table 18: Sys_Time Packed Record : 48 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Seconds	Interfaces.Unsigned_32	0 to 4294967295	32	0	31
Subseconds	Interfaces.Unsigned_16	0 to 65535	16	32	47

Field Descriptions:

- **Seconds** - The number of seconds elapsed since epoch.
- **Subseconds** - The number of 1/(2^16) sub-seconds.

Tick.T:

The tick datatype used for periodic scheduling. Included in this type is the Time associated with a tick and a count.

Table 19: Tick Packed Record : 80 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time	Sys_Time.T	-	48	0	47
Count	Interfaces. Unsigned_32	0 to 4294967295	32	48	79

Field Descriptions:

- **Time** - The timestamp associated with the tick.
- **Count** - The cycle number of the tick.

5.3 Enumerations

The following section outlines any enumerations used in the component.

Data_Product_Enums.Fetch_Status.E:

This status denotes whether a data product fetch was successful.

Table 20: Fetch_Status Literals:

Name	Value	Description
Success	0	The data product was returned successfully.
Not_Available	1	No data product is yet available for the provided id.
Id_Out_Of_Range	2	The data product id was out of range.

Parameter_Enums.Parameter_Operation_Type.E:

This enumeration lists the different parameter operations that can be performed.

Table 21: Parameter_Operation_Type Literals:

Name	Value	Description
Stage	0	Stage the parameter.
Update	1	All parameters are staged, it is ok to update all parameters now.
Fetch	2	Fetch the parameter.
Validate	3	Validate all the parameters.
Discard	4	Discard any parameters staged by this table that have not been updated.

Parameter_Enums.Parameter_Update_Status.E:

This status enumeration provides information on the success/failure of a parameter operation.

Table 22: Parameter_Update_Status Literals:

Name	Value	Description
Success	0	Parameter was successfully staged.
Id_Error	1	Parameter id was not valid.
Validation_Error	2	Parameter values were not successfully validated.

Length_Error	3	Parameter length was not correct.

Triad_Enums.N3_Axis.E:

Inertial z axis direction used as the fallback constraint axis when the sun direction is nearly parallel to the requested thrust direction. Values match the C N3Axis_c enumeration.

Table 23: N3_Axis Literals:

Name	Value	Description
Plus_Z_Hat_N	0	The positive inertial z axis.
Minus_Z_Hat_N	1	The negative inertial z axis.